

Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ MI/E	First Test - Modeule F	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **6 Marks**

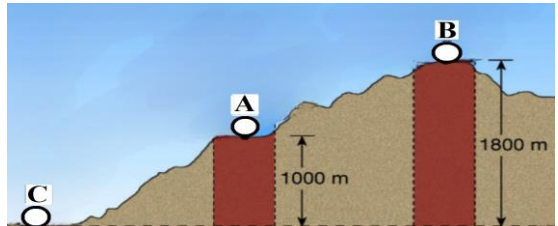
1) Pascal principle used to magnify a force		
2) A suck pressure is a negative gauge pressure		
3) The electric flux decreases as the area increases		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) Which of the following has negative charge:				
a) Electron	b) Proton	c) Neutron	d) Proton & electron	e) None of these

2) The two spheres come into contact and then separate. The charge on sphere B is:				
A -3 C B 3 C				
a) 1.5 C	b) 0 C	c) - 3 C	d) 3 C	e) None of these

3) The force 32 N. If a distance changes from 2 m into 8 m. The force is:				
a) 32 N	b) 0.5 N	c) 16 N	d) 2 N	e) None of these

4) From the figure shown below. The pressures				
				
a) $P_A > P_A$	b) $P_C > P_A$	c) $P_C = P_A$	d) $P_A < P_B$	e) None of these

5) 25 N object displaces 10 N of water. The Buoyant force is:				
a) 10 N	b) 25 N	c) 15 N	d) 0 N	e) None of these

6) An object is held immersed in water and then released to move up:				
a) $\rho_f < \rho_o$	b) $\rho_o = \rho_f$	c) $\rho_f > \rho_o$	d) $\rho_o > \rho_f$	e) None of these

7) Object with $4\text{ m} \times 2\text{ m} \times 4\text{ m}$ having a density of $0.1x\text{ kg/m}^3$. The pressure in Pa on the ground is:				
a) $3x$	b) $0.125x$	c) $30x$	d) $4x$	e) None of these

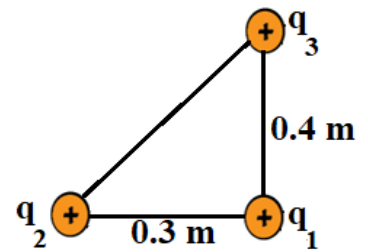
Section III: Answer all the following by selecting the correct choice **20 Marks**
Full and clear steps are required

Constant		
Electron: $m = 9.11 \times 10^{-31}\text{ kg}$	$K_e = 9 \times 10^9\text{ N.m}^2/\text{C}^2$	Gravity: $g = 10\text{ m/s}^2$
Proton: $m = 1.67 \times 10^{-27}\text{ kg}$	$G = 6.67 \times 10^{-11}\text{ N.m}^2/\text{kg}^2$	Water: $\rho_f = 1000\text{ kg/m}^3$

1) A room has dimensions of 3 m and 3 m and a height of 4 m. If the air density is 1.21 kg/m^3 . The weight of the air in the room is:

a) 217.8 N	b) 290.4 N	c) 320.7 N	d) 435.6 N	e) None of these
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2) The total electric force on q_3 is: $q_1 = 2 \mu\text{C}$, $q_2 = 1 \mu\text{C}$ and $q_3 = 1 \mu\text{C}$



a) 0.26 N	b) 0.37 N	c) 0.71 N	d) 0.14 N	e) None of these
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3) Two objects having $m_1 = 2 \text{ kg}$ and $m_2 = 1 \text{ kg}$, separated by 20 cm. The gravitational force is:

a) $66.7 \times 10^{-11} \text{ N}$	b) $333.5 \times 10^{-11} \text{ N}$	c) $88.9 \times 10^{-11} \text{ N}$	d) $667 \times 10^{-11} \text{ N}$	e) None of these
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4) Two opposite charges one is 4 times the other separated by distance r , if a ratio of charge to a distance r is $2 \mu\text{C/m}$. The electric force on $3q$ placed at distance r from a point where $E = 0$ is:

a) 1.33 N	b) 0 N	c) 6.3 N	d) 0.11 N	e) None of these
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5) 400 N object in air displaces 220 N when it is fully immersed in a fluid of $\rho_f = 1500 \text{ kg/m}^3$. The volume of the metal is:

a) 0.0187 m^3	b) 0.0337 m^3	c) 0.008 m^3	d) 0.0147 m^3	e) None of these
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